Adversarial Attacks

Robust Deep Learning (Seminar)

Bjarne Thal   Marius Wrobel

Dept. of Computer Science, RWTH Aachen University, Germany
{bjarne.thal, marius.wrobel}@rwth-aachen.de

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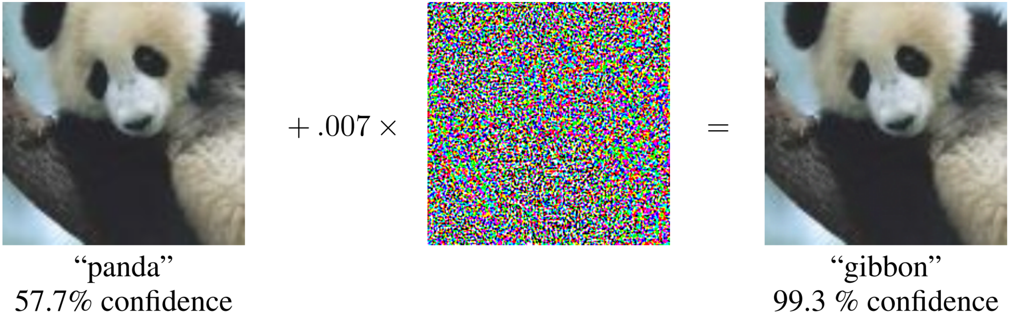
Which one is the gibbon?

Which one is the gibbon?

panda

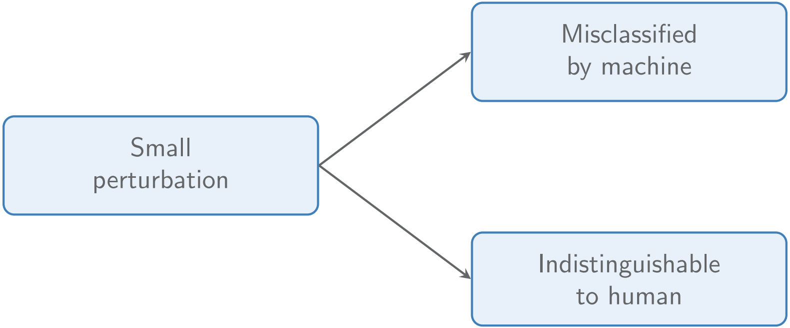
gibbon

Adversarial Examples



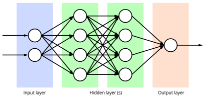
I. J. Goodfellow, Shlens, and Szegedy 2014

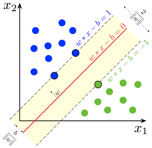
Adversarial Examples

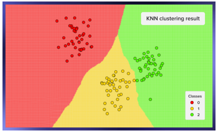


Transferability

- Attacks transfer across architectures, training data, and more

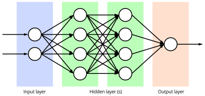


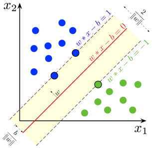


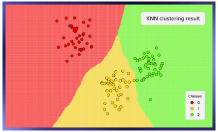


Papernot, McDaniel, and I. Goodfellow 2016

Transferability







- Attacks transfer across architectures, training data, and more

$\rightsquigarrow$ Adv. ex. are properties of a problem itself

Papernot, McDaniel, and I. Goodfellow 2016

Why this matters

- **Fundamental blind spot** while integrated AI systems require high reliability

Images: medicalimaging.na; thetravelimages.com

Outline

Outline

1. Evasion attacks
2. Poisoning attacks
3. Discussion

Taxonomy & Terminology

Attacker knowledge	Attack stage	Attack goal
White-box	Evasion (at test time)	Targeted (specific wrong label)
vs.	vs.	vs.
Black-box	Poisoning (at training time)	Untargeted (any wrong label)

L-BFGS

L-BFGS

Szegedy et al. 2013

White-box · Evasion · Targeted

$$\min_{\delta} \|\delta\|_2 \qquad \text{subject to} \qquad f(x+\delta) = t$$

- f has no gradient — it returns a discrete class label, $f(x+\delta) = t$ is a hard, non-differentiable condition
- Derivative-based optimisers such as L-BFGS need a gradient to move at all

$\leadsto$ Reformulate

L-BFGS

Szegedy et al. 2013

White-box · Evasion · Targeted

$$\min_{\delta} \; c\,\|\delta\|_2 + L_f(x+\delta,\; t)$$

Significance

- First method to **reliably generate** adversarial examples
- Computationally expensive — far too slow to be practical

$\leadsto$ It *proved a point* rather than providing a usable attack

Fast Gradient Sign Method (FGSM)

Fast Gradient Sign Method

I. J. Goodfellow, Shlens, and Szegedy 2014

White-box · Evasion · Untargeted

- Previously: Adversarial examples assumed to be caused by 'complex' behaviour
- Goodfellow et al.: Linear models have adv. ex. as well

$\rightsquigarrow$ Models may be too linear

Do Neural Networks behave linearly?

I. J. Goodfellow, Shlens, and Szegedy 2014

...

Fast Gradient Sign Method

I. J. Goodfellow, Shlens, and Szegedy 2014

White-box · Evasion · Untargeted

- Linearise the loss function:

$$\arg\max_{||\delta||_\infty \leq \epsilon} L_f(x+\delta, y) = \arg\max_{||\delta||_\infty \leq \epsilon} L_f(x, y) + \nabla_x L_f(x, y)^T \delta$$

- Maximised by $\delta = \epsilon \cdot sign(\nabla_x L_f(x, y))$

$\rightsquigarrow$ Single step in gradient direction

Fast Gradient Sign Method - Example

I. J. Goodfellow, Shlens, and Szegedy 2014

White-box · Evasion · Untargeted

placeholder: step graph here

[Results/Significance]

I. J. Goodfellow, Shlens, and Szegedy 2014

White-box · Evasion · Untargeted

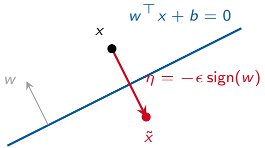
- Maximised by $\delta = \epsilon \cdot sign(\nabla_x L_f(x, y))$

$\rightsquigarrow$ Single step in gradient direction

High-dimensional linearity

I. J. Goodfellow, Shlens, and Szegedy 2014

- Affine case first: perturbation shifts activation by $w^\top \eta$



$$w^\top \tilde{x} = w^\top (x + \eta)$$
$$= w^\top x + w^\top \eta$$

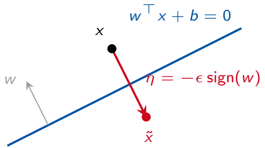
$$w^\top \eta = \epsilon \sum_{i=1}^{n} |w_i| = \epsilon\, m\, n$$

n dimensions, mean weight magnitude m

High-dimensional linearity

I. J. Goodfellow, Shlens, and Szegedy 2014

- Affine case first: perturbation shifts activation by $w^\top \eta$
- Best strategy: η aligned with the weights



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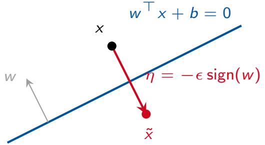
- Score change scales *linearly* with number of dimensions w.r.t. input change



$$
\begin{aligned}
w^\top \tilde{x} &= w^\top (x + \eta) \\
&= w^\top x + w^\top \eta
\end{aligned}
$$

$$
w^\top \eta = \epsilon \sum_{i=1}^{n} |w_i| = \epsilon\, m\, n
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n dimensions, mean weight magnitude m

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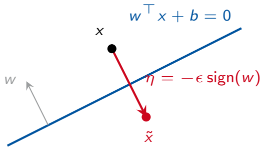
- Score change scales *linearly* with number of dimensions w.r.t. input change

$\leadsto$ Many imperceptible changes accumulate into one large shift



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n dimensions, mean weight magnitude m

From linear models to deep networks

I. J. Goodfellow, Shlens, and Szegedy 2014

- Most DNN architectures behave *locally linearly* for ease of optimisation

ReLU

LSTM

sigmoid near zero

maxout

linear-softmax

From linear models to deep networks

I. J. Goodfellow, Shlens, and Szegedy 2014

- Most DNN architectures behave *locally linearly* for ease of optimisation

$\rightsquigarrow$ Within ϵ-ball, the linear argument carries over

ReLU

LSTM

sigmoid near zero

maxout

linear-softmax

Projected Gradient Descent (PGD)

PGD
Madry et al. 2017

White-box · Evasion · ???

General formulation of adversarial training [...! incl. defenders part?]. Adversary solves

$$\max_{\delta \in S} L_f(x + \delta, y)$$

- FGSM: Use $S = \{\delta \mid \|\delta\|_\infty \leq \epsilon\}$
- Use projected gradient descent to optimise

$\leadsto$ Models may be too linear

PGD - a general first-order attack?
Madry et al. 2017

White-box · Evasion · ???

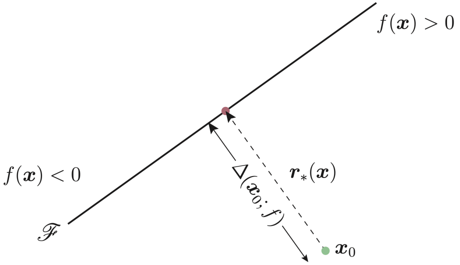
DeepFool

DeepFool

Moosavi-Dezfooli, Fawzi, and Frossard 2016 — the affine, binary case

White-box · Evasion · Untargeted

- Decision boundary is a *hyperplane*
- Minimal perturbation is *orthogonal projection* of x onto $\mathcal{F}$ $\rightarrow$ closed form

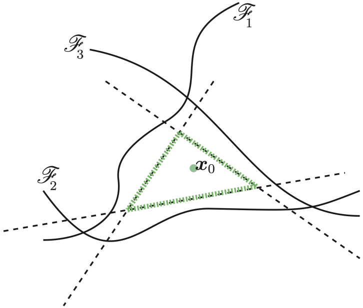


DeepFool

White-box · Evasion · Untargeted

Moosavi-Dezfooli, Fawzi, and Frossard 2016 — the general, multiclass case

- **Non-linear:** iterative → linearise the score function at current point (first-order Taylor), project onto that hyperplane, repeat until label flips
- **Multiclass:** Linearise *every* class boundary, project onto the **closest**



Significance

- **Geometric reading**: robustness as distance to the decision boundary

Transfer attacks

Transfer attacks

- Black-box attack pattern
- Adversary creates and trains a substitute model for the target
- Adversarial examples for substitute model are found (white-box!)
- These can be used to attack the target

Jacobian-Based Dataset Augmentation

Papernot, McDaniel, I. Goodfellow, et al. 2017

Black-box · Evasion · ???

- Assumption: No knowledge of model parameters, dataset
  Access to somewhat representative inputs is granted (e.g. natural images)
- How do we construct a suitable substitute?
- Construct an artificial dataset!

Jacobian-Based Dataset Augmentation

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Black-box · Evasion · ???

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- How do we construct a suitable substitute? (similar decision boundaries)
- Construct an artificial dataset!

Jacobian-Based Dataset Augmentation

Papernot, McDaniel, I. Goodfellow, et al. 2017

- Use target model as an oracle to label samples
- Strategy for sample selection is necessary
- Idea: iteratively add meaningful samples to dataset
- Use Jacobian for a heuristic

Heuristic for meaningful directions

Papernot, McDaniel, I. Goodfellow, et al. 2017

$$x = x + \alpha \cdot sign(J_f(x)[o(x)])$$

- $J_f(x)$: total derivative
- $J_f(x)[o(x)]$: total derivative, column of the correct label

$\rightsquigarrow$ We move into the class region (according to the substitute)!

Jacobian-Based Dataset Augmentation

Papernot, McDaniel, I. Goodfellow, et al. 2017

Black-box · Evasion · ???

- 1) Collect samples somewhat related to the input domain
- 2) Choose architecture for substitute
- 3) Target model labels current samples
- 4) Train substitute
- 5) Expand dataset: $S_{N+1} := \{x = x + \alpha \cdot sign(J_f(x)[o(x)]) \mid x \in S_N\} \cup S_N$
- Repeat (3) onwards

White-box attacks to improve transferability

- Adversarial examples can differ in transferability!
- MI-FGSM: adds momentum to FGSM
- Ensemble attacks

ZOO

ZOO: Zeroth order black-box attacks

Black-box · Evasion · Targ./Untarg.

- Black-box attack without substitute model
- No quality loss due to transfer
- Assumptions: Adversary allowed to query, gets confidence scores

Coordinate-wise pseudo gradient

Black-box · Evasion · Targ./Untarg.

$$\hat{g}_i := \frac{\partial l(x)}{\partial x_i} \approx \frac{l(x + he_i) - l(x - he_i)}{2h}$$

- l loss function based on confidence scores
- Two queries per coordinate, full gradient infeasible
- -¿ Coordinate descent using first and/or second-order method

Proposed loss function

Black-box · Evasion · Targ./Untarg.

- Reminder: Carlini-Wagner loss: $\ell(x') = \max\big(\max_{i\neq t} z_i(x') - z_t(x'),\ 0\big)$
- Targeted ZOO: $l(x,t) = \max\{\max_{i\neq t}\{\log[f(x)]_i\} - \log[f(x)]_t, -\kappa\}$
- Untargeted ZOO: $l(x) = \max\{\log[f(x)]_y - \max_{i\neq y}\{\log[f(x)]_i\}, -\kappa\}$
- *log* to estimate logits

Outline

1. Evasion attacks
2. Poisoning attacks
3. Discussion

Hidden Trigger Backdoor Attack

Hidden Trigger Backdoor Attack

Black-box · ??? ·

- Goal: get ML model to misclassify any sample with trigger pattern into target class t
- Similar threat model to Poison Frogs
- Restriction: adversary cannot choose labels
- -¿ "Clean-label attack"

Hiding the trigger

Black-box · Evasion · Targ./Untarg.

- Sample instances s b of target class t
- Apply trigger to source: patched source $\tilde{s}$
- Poison instances:

$$\arg\min_{x} ||g(x) - g(\tilde{s})||_2^2 \qquad \textbf{s.t.} \quad ||x - b||_\infty < \epsilon$$

- g: feature representation (second-to-last layer output)
- Keep x close to $\tilde{s}$ in feature space, but close to b in pixel space

Outline

Explaining adversarial examples

2013 **Szegedy et al.** — quirks of non-linearity

Explaining adversarial examples

2013 **Szegedy et al.** — quirks of non-linearity

2014 **Goodfellow et al.** — high-dimensional linearity

Explaining adversarial examples

2013 **Szegedy et al.** — quirks of non-linearity

2014 **Goodfellow et al.** — high-dimensional linearity

2019 **Ilyas et al.** — predictive but non-robust features

Predictive and robust features
Ilyas et al. 2019

Predictive

Correlated with the label in expectation.

Predictive and robust features

Ilyas et al. 2019

Predictive

Correlated with the label in expectation.

Robust

Stays correlated *under* adversarial perturbation.

Predictive and robust features

Ilyas et al. 2019

Predictive

Correlated with the label in expectation.

Robust

Stays correlated *under* adversarial perturbation.

$\leadsto$ **Predictive but non-robust:** Standard training happily relies on such features

Experiment: "Mislabeled" training dataset
Ilyas et al. 2019

Real-life adversarial attacks

- Real confirmed malicious attacks are surprisingly rare
- Two candidate explanations:
  1. They slip through unnoticed, logged as ordinary misclassifications
  2. They are simply of little use to actual attackers

Adversarial attacks as a subject of study

- Not really significant security threat

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- Mainly *epistemic significance*:
  - Not: What can an attacker do?
  - But: What do they reveal about the targetted system?

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Before

- Accuracy as near-single metric
- Robustness assumed implicitly

Adversarial attacks as a subject of study

- Not really significant security threat
- Mainly *epistemic significance*:
  - Not: What can an attacker do?
  - But: What do they reveal about the targetted system?

Before

- Accuracy as near-single metric
- Robustness assumed implicitly

Afterwards

- High accuracy $\neq$ human-like understanding
- Robustness as property of interest

Key takeaways

- Imperceptible perturbation $\rightarrow$ wrong prediction
- Transferable across models, architectures, training data
- Categories: white/black-box · evasion/poisoning · targeted/untargeted
- Phenomenon still not fully explained
- Marginal security threat so far but epistemic relevance

Thank you for your attention!

Questions?

References I

Goodfellow, Ian J, Jonathon Shlens, and Christian Szegedy (2014). "Explaining and harnessing adversarial examples". In: *arXiv preprint arXiv:1412.6572*.

Papernot, Nicolas, Patrick McDaniel, and Ian Goodfellow (2016). "Transferability in machine learning: from phenomena to black-box attacks using adversarial samples". In: *arXiv preprint arXiv:1605.07277*.

Szegedy, Christian et al. (2013). "Intriguing properties of neural networks". In: *arXiv preprint arXiv:1312.6199*.

Madry, Aleksander et al. (2017). "Towards deep learning models resistant to adversarial attacks". In: *arXiv preprint arXiv:1706.06083*.

Moosavi-Dezfooli, Seyed-Mohsen, Alhussein Fawzi, and Pascal Frossard (2016). "Deepfool: a simple and accurate method to fool deep neural networks". In: *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2574–2582.

Papernot, Nicolas, Patrick McDaniel, Ian Goodfellow, et al. (2017). "Practical black-box attacks against machine learning". In: *Proceedings of the 2017 ACM on Asia conference on computer and communications security*, pp. 506–519.

Ilyas, Andrew et al. (2019). "Adversarial examples are not bugs, they are features". In: *Advances in Neural Information Processing Systems* 32.

References II

Carlini, Nicholas and David Wagner (2017). "Towards evaluating the robustness of neural networks". In: *2017 ieee symposium on security and privacy (sp)*. Ieee, pp. 39–57.

Sharif, Mahmood et al. (2016). "Accessorize to a Crime: Real and Stealthy Attacks on State-of-the-Art Face Recognition". In: *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, pp. 1528–1540.

Xu, Kaidi et al. (2020). "Adversarial T-Shirt! Evading Person Detectors in a Physical World". In: *Computer Vision – ECCV 2020*. Springer, pp. 665–681. DOI: 10.1007/978-3-030-58558-7_39.

Eykholt, Kevin et al. (2018). "Robust Physical-World Attacks on Deep Learning Visual Classification". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1625–1634.

Athalye, Anish et al. (2018). "Synthesizing robust adversarial examples". In: *International conference on machine learning*. PMLR, pp. 284–293.

Backup Slides

Why that works
Ilyas et al. 2019

	Robust features point at	Non-robust features point at	Label	Image
Original clean training image (in $\mathcal{D}$)	dog	dog	dog	
Perturbed training image (in $\widehat{\mathcal{D}}_{NR}$)	dog *(unaffected by perturbation)*	cat *(through adversarial perturbation)*	cat	
Original clean test image (in $\mathcal{D}$)	cat	cat	**cat** (predicted)	

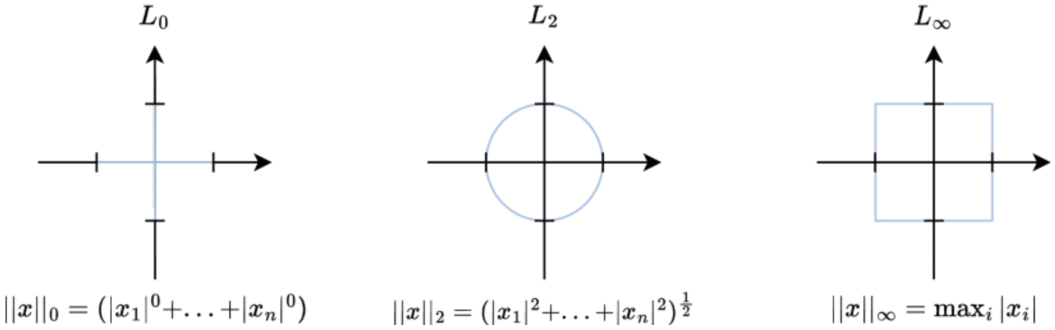
What is an Adversarial Example?

Definition (Adversarial Example)

An *adversarial example* is an **input to a machine learning model** that has been intentionally perturbed, typically **in a subtle and often imperceptible manner**, such that the model **produces an incorrect prediction** while the semantic content of the input remains unchanged.

- **Machine learning models**: No quirk of deep networks, but a general property of machine learners
- **Subtle and imperceptible perturbations**: mathematically small under a chosen norm / visually imperceptible to humans
- **Produces an incorrect predition**: The machine learner is fooled where the human would not have been

Measuring “small”: the ℓ_p norms



Norm defines the **perturbation budget** ϵ.

Carlini & Wagner

Carlini and Wagner 2017

White-box · Evasion · Targeted

$$\underbrace{\min_{\delta}\ c\,\|\delta\|_2 + L_f(x+\delta,\ t)}_{\text{L-BFGS}} \qquad \longrightarrow \qquad \underbrace{\min_{\delta}\ \|\delta\|_p + c \cdot \ell(x+\delta)}_{\text{C\&W}}$$

- $\ell(x') = \max\big(\max_{i \neq t} z_i(x') - z_t(x'),\ 0\big)$
- Built so that $\ell(x') \leq 0 \iff$ input is classified as $t \rightarrow$ faithfully encodes the attack goal
- c decided through **binary search**

Carlini & Wagner

Carlini and Wagner 2017

White-box · Evasion · Targeted

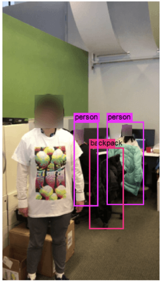
$$x' = \frac{1}{2}\big(\tanh(w) + 1\big), \qquad w \in \mathbb{R}^m$$

- Mapping any $w \in \mathbb{R}^m$ into $(0,1)^m$ $\rightarrow$ every w decodes to a **valid image** $\rightarrow$ dissolving box-constraint
- Optimise w **unconstrained** with Adam optimiser

Significance

- Strongest attack of its time across $\ell_0/\ell_2/\ell_\infty$
- Broke *defensive distillation*, exposing it as *gradient masking*

Physical adversarial attacks



Sharif et al. 2016 · Xu et al. 2020

Eykholt et al. 2018

Athalye et al. 2018

Poison Frogs

Black-box · Evasion · Targ./Untarg.

- Goal: get ML model to misclassify one specific sample v into target class t
- Assumption: model based on pretrained weights known to the attacker, access to data distribution
- Restriction: adversary cannot choose labels
- -¿ "Clean-label attack"

Poison Frogs: poison instances

Black-box · Evasion · Targ./Untarg.

- Sample instance b of target class t
- Poison instance(s):

$$p = \arg\min_{x} ||g(x) - g(v)||_2^2 + \lambda ||x - b||_2^2$$

- g: feature representation (second-to-last layer output)
- Keep x close to v in feature space, but close to b in pixel space